

Fall 2026 EE 102: In-class Activity (Sep 16, 2026)

Bigger Energy: Electric Vehicles Edition

You are driving an electric vehicle (EV) on a one-hour trip from point A to point B. There are many ways you can do this trip — which one uses the least energy?

Each team is assigned a simple signal trace $x(t)$ that represents how much battery is used up in an electric vehicle over $[0, 1]$ hour for a given scenario (assume arbitrary units for the math to work out). Your task is to compute the total energy E (in kWh) to determine who has the biggest energy consumption and predict your rank (1 = least energy spent). Among your team, you should confirm that everyone has the same answer and come to a consensus of the rank. Then, you will be asked to announce your energy and your guess of where you stand.

Key formula (use t in hours):

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt.$$

Team Prompts (pick one per team)

All signals are zero outside the intervals shown.

Team 1: Flat road, stop, flat road

$$x_1(t) = \begin{cases} 12, & 0 \leq t < \frac{1}{3} \\ 0, & \frac{1}{3} \leq t < \frac{2}{3} \\ 12, & \frac{2}{3} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 1: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 2: Uphill, stop, downhill

$$x_2(t) = \begin{cases} 18, & 0 \leq t < \frac{1}{3} \quad (\text{uphill}) \\ 0, & \frac{1}{3} \leq t < \frac{2}{3} \quad (\text{stop}) \\ 6, & \frac{2}{3} \leq t \leq 1 \quad (\text{downhill}) \\ 0, & \text{otherwise} \end{cases}$$

Team 2: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 3: Rapid cruise (less efficient) to destination

$$x_3(t) = \begin{cases} 13.5, & 0 \leq t < \frac{2}{3} \\ 0, & \frac{2}{3} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 3: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 4: Dynamic speeding (ramp up, then ramp down)

Ramp up such that battery uses from 8 to 16 over the first 0.5 h, then ramp down to 8 over the next 0.5 h.

$$x_4(t) = \begin{cases} 8 + 16t, & 0 \leq t < \frac{1}{2} \\ 24 - 16t, & \frac{1}{2} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 4: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 5: Stop-and-go traffic (three bursts)

Scenario where $x(t) = 18$ for three 10 min segments, with 10 min stops between.

$$x_5(t) = \begin{cases} 18, & 0 \leq t < \frac{1}{6}, \frac{1}{3} \leq t < \frac{1}{2}, \frac{2}{3} \leq t < \frac{5}{6} \\ 0, & \text{else on } [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

Team 5: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 6: Eco mode (slow and steady)

$$x_6(t) = \begin{cases} 9, & 0 \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 6: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)

Team 7: Late starter, speeding later

$$x_7(t) = \begin{cases} 0, & 0 \leq t < \frac{1}{2} \\ 16, & \frac{1}{2} \leq t \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Team 7: Record your results

Energy E = _____ kWh Predicted rank = _____ (/8)